

MATHEMATICAL ANALYSIS / ACHIEVEMENT SETS

THE BASE-4 CANTORVAL

An explicit candidate resolution of the Second Jones Problem



$$x_{2n-1} = \frac{8n + 17}{4^n}, \quad x_{2n} = \frac{4n + 18}{4^n}, \quad \lim_{k \rightarrow \infty} \frac{x_{k+1}}{x_k} = \frac{1}{2}$$

PAIR DIGITS FORM A COMPLETE RESIDUE SYSTEM MODULO 4



AN EXACT GAP AT EVERY SCALE

$$x_{2n} - r_{2n} = 4^{-n}$$

An Explicit Cantorval Achievement Set with Convergent Consecutive-Term Ratio

A candidate affirmative solution to the Second Jones Problem

Metriq PRISM Laboratory

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Research status

This manuscript presents an explicit candidate resolution of a published open problem. It has not undergone external peer review. A December 2025 survey lists the Second Jones Problem as open [2, Problem 4]. A targeted literature search completed on 18 July 2026 did not identify a later published resolution or the explicit sequence below; that search does not establish priority. The proof is exact and non-computational, but independent specialist verification is required before the result should be treated as established.

Abstract

For a summable positive sequence $x = (x_k)$, its achievement set is the compact set of all subsums

$$\mathcal{E}(x) = \left\{ \sum_{k=1}^{\infty} \varepsilon_k x_k : \varepsilon_k \in \{0, 1\} \right\}.$$

The Second Jones Problem asks whether there is a sequence for which the consecutive-term ratio has a limit while the achievement set is a Cantorval. We give the explicit construction

$$x_{2n-1} = \frac{8n+17}{4^n}, \quad x_{2n} = \frac{4n+18}{4^n} \quad (n \geq 1).$$

We prove that x is strictly decreasing and summable, that $\lim_{k \rightarrow \infty} x_{k+1}/x_k = 1/2$, that $\mathcal{E}(x)$ has nonempty interior, and that $\mathcal{E}(x)$ has infinitely many gaps. The interior argument is driven by the fact that the four pair-digits form a complete residue system modulo 4 at every scale. The gaps follow from an exact tail identity. The Guthrie–Nymann classification theorem, with the corrected proof of Nymann and Sáenz, then implies that $\mathcal{E}(x)$ is a Cantorval. Subject to independent verification, this gives an affirmative answer to the Second Jones Problem.

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MAIN CONSTRUCTION

$$x_{2n-1} = \frac{8n+17}{4^n}, \quad x_{2n} = \frac{4n+18}{4^n}, \quad \sum_{k=1}^{\infty} x_k = 17,$$

$$\lim_{k \rightarrow \infty} \frac{x_{k+1}}{x_k} = \frac{1}{2}, \quad \text{int } \mathcal{E}(x) \neq \emptyset, \quad \left(\frac{4n+17}{4^n}, \frac{4n+18}{4^n} \right) \cap \mathcal{E}(x) = \emptyset.$$

1 Introduction

Let $x = (x_k)_{k \geq 1}$ be an absolutely summable real sequence. Its *achievement set*, also called its set of subsums, is

$$\mathcal{E}(x) := \left\{ \sum_{k=1}^{\infty} \varepsilon_k x_k : (\varepsilon_k)_{k \geq 1} \in \{0, 1\}^{\mathbb{N}} \right\}. \quad (1)$$

The systematic study of such sets goes back to Kakeya [5]. For an infinite positive summable sequence, $\mathcal{E}(x)$ is compact and perfect. The one-dimensional classification theorem of Guthrie and Nymann [3], whose proof was corrected by Nymann and Sáenz [8], states that an achievement set belongs to one of the following topological alternatives: a finite set, a finite union of closed bounded intervals, a Cantor set, or a Cantorval.

Jones [4, p. 519] asked whether a sequence can simultaneously have a convergent adjacent-term ratio and a Cantorval achievement set. A recent survey records the question as follows [2, Problem 4]:

Does there exist a sequence (x_n) such that $\lim_{n \rightarrow \infty} x_{n+1}/x_n$ exists and $\mathcal{E}(x_n)$ is a Cantorval?

Known Cantorval constructions are often multigeometric, and their periodic block structure usually prevents the full consecutive-term ratio from converging unless the block degenerates. The construction in this paper replaces fixed block coefficients with slowly varying linear coefficients. This preserves the base-4 digit mechanism that creates interior while making both types of adjacent ratios converge to the same value.

The proof has two independent geometric components. First, the four possible subsums from the n th pair represent all residue classes modulo 4. A finite-union category argument then forces $\mathcal{E}(x)$ to contain an interval. Second, the even term x_{2n} exceeds the entire remaining tail by exactly 4^{-n} , producing a gap at every scale. These two facts exclude both the Cantor-set and multi-interval alternatives in the classification theorem.

2 Preliminaries

Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$. For a positive summable sequence x , write

$$r_k := \sum_{j=k+1}^{\infty} x_j \quad (2)$$

for the remainder after the k th term.

Definition 2.1. A *multi-interval set* is a finite union of bounded closed intervals. A *Cantor set* is a set homeomorphic to the classical ternary Cantor set. A *Cantorval* is the remaining topological alternative in the Guthrie–Nymann achievement-set classification after finite sets, multi-interval sets, and Cantor sets are excluded; equivalently, it is homeomorphic to the model achievement set generated by

$$\frac{3}{4}, \frac{2}{4}, \frac{3}{4^2}, \frac{2}{4^2}, \dots$$

See [3, 8, 6].

We shall use only the following consequence of the classification theorem.

Proposition 2.2 (Classification consequence). *Let x be an infinite positive summable sequence. If $\mathcal{E}(x)$ has nonempty interior and infinitely many gaps, then $\mathcal{E}(x)$ is a Cantorval.*

Proof. Nonempty interior rules out a Cantor set. Infinitely many gaps rule out a finite union of closed intervals. The set is not finite because x has infinitely many positive terms. The classification theorem leaves only the Cantorval alternative. $\square$

3 The construction and main result

Define, for every $n \geq 1$,

$$a_n := \frac{8n + 17}{4^n}, \quad b_n := \frac{4n + 18}{4^n}, \quad (3)$$

and interlace the two sequences by

$$x_{2n-1} = a_n, \quad x_{2n} = b_n. \quad (4)$$

The first terms are

$$\frac{25}{4}, \frac{22}{4}, \frac{33}{16}, \frac{26}{16}, \frac{41}{64}, \frac{30}{64}, \dots$$

Theorem 3.1 (Main theorem). *The sequence defined by (3)–(4) is positive, strictly decreasing, and satisfies*

$$\sum_{k=1}^{\infty} x_k = 17, \quad \lim_{k \rightarrow \infty} \frac{x_{k+1}}{x_k} = \frac{1}{2}.$$

Its achievement set $\mathcal{E}(x)$ has nonempty interior and infinitely many gaps. Consequently, $\mathcal{E}(x)$ is a Cantorval.

The constants in (3) meet three requirements at once:

Requirement	Mechanism	Exact identity
Convergent adjacent ratios	Linear leading coefficients $8n$ and $4n$, combined with the block factor 4^{-n}	Both adjacent-ratio subsequences tend to $1/2$
Nonempty interior	Pair digits exhaust all residues modulo 4	$0, b_n 4^n, a_n 4^n, (a_n + b_n) 4^n \equiv 0, 2, 1, 3 \pmod{4}$
Infinitely many gaps	Each even term is slightly larger than the following tail	$x_{2n} - r_{2n} = 4^{-n}$

For geometric orientation, let S_N be the set of subsums of the first N pairs and set

$$U_N := \bigcup_{s \in S_N} [s, s + r_{2N}].$$

Then $\mathcal{E}(x) \subseteq U_N$. The first five exact outer approximations are shown in Figure 1; their recurrent gaps are visible even though the proof of interior below is arithmetic rather than graphical.

4 Elementary properties and the ratio limit

Lemma 4.1. *The sequence x is positive and strictly decreasing, and its total sum is 17.*

Proof. Positivity is immediate. Within the n th pair,

$$x_{2n-1} - x_{2n} = \frac{(8n + 17) - (4n + 18)}{4^n} = \frac{4n - 1}{4^n} > 0.$$



Figure 1: Exact interval outer approximations $U_1, \dots, U_5$, normalized horizontally to the total sum 17. Each row is computed from all subsums of the first N pairs enlarged by the full remaining tail.

Between successive pairs,

$$\begin{aligned} x_{2n} - x_{2n+1} &= \frac{4n + 18}{4^n} - \frac{8(n + 1) + 17}{4^{n+1}} \\ &= \frac{16n + 72 - (8n + 25)}{4^{n+1}} \\ &= \frac{8n + 47}{4^{n+1}} > 0. \end{aligned}$$

Thus $x_1 > x_2 > x_3 > \dots > 0$.

The sum of the n th pair is

$$x_{2n-1} + x_{2n} = \frac{12n + 35}{4^n}.$$

Using

$$\sum_{n=1}^{\infty} \frac{n}{4^n} = \frac{4}{9}, \quad \sum_{n=1}^{\infty} \frac{1}{4^n} = \frac{1}{3},$$

we obtain

$$\sum_{k=1}^{\infty} x_k = 12 \left(\frac{4}{9} \right) + 35 \left(\frac{1}{3} \right) = 17.$$

□

Proposition 4.2. *The consecutive-term ratio converges to $1/2$.*

Proof. For adjacent terms within one pair,

$$\frac{x_{2n}}{x_{2n-1}} = \frac{4n + 18}{8n + 17} \rightarrow \frac{1}{2}.$$

For the transition from one pair to the next,

$$\frac{x_{2n+1}}{x_{2n}} = \frac{(8n + 25)/4^{n+1}}{(4n + 18)/4^n} = \frac{8n + 25}{16n + 72} \rightarrow \frac{1}{2}.$$

Every adjacent ratio belongs to one of these two subsequences, so the full sequence converges to $1/2$. □

5 A complete residue system forces interior

Group the terms into pairs. From the n th pair, a subsum contributes one of the four quantities

$$0, \quad \frac{4n + 18}{4^n}, \quad \frac{8n + 17}{4^n}, \quad \frac{12n + 35}{4^n}.$$

Define the integer digit set

$$D_n := \{0, 4n + 18, 8n + 17, 12n + 35\}. \quad (5)$$

Then

$$\mathcal{E}(x) = \left\{ \sum_{n=1}^{\infty} \frac{d_n}{4^n} : d_n \in D_n \right\}. \quad (6)$$

Modulo 4, the elements of D_n are respectively

$$0, \quad 2, \quad 1, \quad 3.$$

Hence D_n contains exactly one representative of each residue class modulo 4.

Lemma 5.1 (Mixed-radix residue lemma). Fix $N \geq 1$. For $d_n \in D_n$, define

$$P_N(d_1, \dots, d_N) := \sum_{n=1}^N d_n 4^{N-n}.$$

As $(d_1, \dots, d_N)$ ranges over $D_1 \times \dots \times D_N$, the integers $P_N(d_1, \dots, d_N)$ represent every residue class modulo 4^N exactly once.

Proof. There are 4^N digit strings, so it is enough to prove injectivity modulo 4^N . Suppose

$$P_N(d_1, \dots, d_N) \equiv P_N(d'_1, \dots, d'_N) \pmod{4^N}.$$

Reducing modulo 4 gives $d_N \equiv d'_N \pmod{4}$. Because D_N contains one representative of each residue class modulo 4, we have $d_N = d'_N$. Subtract the common last digit and divide the congruence by 4. The same argument now applies modulo 4^{N-1} to the first $N - 1$ digits. Iteration yields $d_n = d'_n$ for every $n \leq N$. $\square$

Proposition 5.2. The achievement set $\mathcal{E}(x)$ has nonempty interior.

Proof. Fix $N \geq 1$ and an integer r with $0 \leq r < 4^N$. By Lemma 5.1, there is a digit string $(d_1, \dots, d_N)$ such that

$$P_N(d_1, \dots, d_N) = r + m4^N$$

for some integer m . Dividing by 4^N gives

$$\sum_{n=1}^N \frac{d_n}{4^n} = m + \frac{r}{4^N}. \quad (7)$$

The left side is a finite subsum, hence an element of $\mathcal{E}(x)$. It is nonnegative and strictly less than the total sum 17. Since $0 \leq r/4^N < 1$, it follows that

$$m \in \{0, 1, \dots, 16\}.$$

Thus

$$\left\{ \frac{r}{4^N} : 0 \leq r < 4^N \right\} \subseteq \bigcup_{m=0}^{16} (\mathcal{E}(x) - m). \quad (8)$$

Taking the union over N , the left side is dense in $[0, 1]$. The set $\mathcal{E}(x)$ is compact: it is the continuous image of the compact product space $\{0, 1\}^{\mathbb{N}}$ under the uniformly convergent subsum map. Hence the finite union

$$F := \bigcup_{m=0}^{16} (\mathcal{E}(x) - m)$$

is closed. Equation (8) therefore implies $[0, 1] \subseteq F$.

Assume for contradiction that $\text{int } \mathcal{E}(x) = \emptyset$. Because $\mathcal{E}(x)$ is closed, it is nowhere dense; every translate $\mathcal{E}(x) - m$ is likewise nowhere dense. A finite union of nowhere-dense sets is nowhere dense and cannot contain the interval $[0, 1]$. This contradicts $[0, 1] \subseteq F$. Therefore $\text{int } \mathcal{E}(x) \neq \emptyset$. $\square$

Remark 5.3. The proof does not locate a particular interval inside $\mathcal{E}(x)$. Instead, the complete residue systems show that finitely many integer translates of $\mathcal{E}(x)$ cover a dense base-4 grid; compactness and category force one translate, and therefore $\mathcal{E}(x)$ itself, to contain an interval.

6 An exact gap at every scale

Lemma 6.1 (Tail identity). For every $n \geq 1$,

$$r_{2n} = \sum_{k=2n+1}^{\infty} x_k = \frac{4n+17}{4^n}. \quad (9)$$

Consequently,

$$x_{2n} - r_{2n} = \frac{1}{4^n} > 0. \quad (10)$$

Proof. Grouping all later terms into pairs gives

$$r_{2n} = \sum_{j=n+1}^{\infty} \frac{12j+35}{4^j}.$$

The geometric-series identities

$$\sum_{j=n+1}^{\infty} \frac{1}{4^j} = \frac{1}{3 \cdot 4^n}, \quad \sum_{j=n+1}^{\infty} \frac{j}{4^j} = \frac{3n+4}{9 \cdot 4^n}$$

yield

$$\begin{aligned} r_{2n} &= 12 \frac{3n+4}{9 \cdot 4^n} + 35 \frac{1}{3 \cdot 4^n} \\ &= \frac{4n+17}{4^n}. \end{aligned}$$

Since $x_{2n} = (4n+18)/4^n$, equation (10) follows. □

Proposition 6.2. For every $n \geq 1$, the open interval

$$G_n := \left(\frac{4n+17}{4^n}, \frac{4n+18}{4^n} \right) = (r_{2n}, x_{2n}) \quad (11)$$

is a gap of $\mathcal{E}(x)$. The gaps G_n are pairwise distinct.

Proof. Both endpoints of G_n belong to $\mathcal{E}(x)$. The left endpoint r_{2n} is obtained by selecting every term after x_{2n} and no earlier term. The right endpoint x_{2n} is obtained by selecting only x_{2n} .

Now take any subsum. If it uses none of the first $2n$ terms, it is at most r_{2n} . If it uses at least one of the first $2n$ terms, then it is at least x_{2n} , because the sequence is decreasing and x_{2n} is the smallest of those first $2n$ terms. Hence no subsum lies strictly between r_{2n} and x_{2n} , proving $G_n \cap \mathcal{E}(x) = \emptyset$.

Finally,

$$x_{2n+2} = \frac{4n+22}{4^{n+1}} < \frac{16n+68}{4^{n+1}} = r_{2n}.$$

Thus G_{n+1} lies strictly to the left of G_n , so the gaps are pairwise distinct. □

7 Classification and resolution of the problem

Proof of theorem 3.1. The elementary properties and total sum are established in Lemma 4.1. The ratio limit is theorem 4.2. The achievement set has nonempty interior by theorem 5.2 and infinitely many gaps by theorem 6.2. Applying theorem 2.2, we conclude that $\mathcal{E}(x)$ is a Cantorval. $\square$

Corollary 7.1 (Normalized form). *Let $y_k = x_k/17$. Then*

$$\sum_{k=1}^{\infty} y_k = 1, \quad \lim_{k \rightarrow \infty} \frac{y_{k+1}}{y_k} = \frac{1}{2},$$

and $\mathcal{E}(y) = \frac{1}{17} \mathcal{E}(x)$ is a Cantorval.

Proof. Positive scaling preserves adjacent ratios and topological type, and scales the total sum and achievement set by the same factor. $\square$

Remark 7.2 (Answer to the Second Jones Problem). The sequence in (3)–(4) has a convergent consecutive-term ratio and a Cantorval achievement set. Accordingly, the theorem gives an affirmative answer to the Second Jones Problem as formulated by Jones [4, p. 519] and restated in [2, Problem 4], subject to independent verification of this manuscript.

8 Discussion

The construction is not multigeometric: the coefficients within each base-4 block vary with n . It nevertheless retains the decisive arithmetic feature of the Guthrie–Nymann model, namely a full set of residues at every base-4 digit position. This suggests a broader design principle.

Suppose a block construction in base q has finite digit sets $D_n \subset \mathbb{N}$ that are complete residue systems modulo q , while all finite partial sums remain in a fixed bounded interval. Then a variant of Lemma 5.1 and Proposition 5.2 can force interior through finitely many translates. If, independently, selected block terms exceed their following tails infinitely often, the same construction may generate infinitely many gaps. Slowly varying block coefficients can then be tuned so that distinct adjacent-ratio subsequences converge to a common limit.

The present paper does not attempt a classification of such variable-block constructions. Two immediate directions are:

1. determine parameter families that preserve the complete-residue and exact-gap identities; and
2. identify the location, measure, and component structure of the interior of $\mathcal{E}(x)$ beyond the existence argument given here.

No numerical approximation is needed for the proof. The only external structural input is the one-dimensional achievement-set classification theorem.

Reproducibility materials

The accompanying source archive contains the complete Lua[®]TeX manuscript, the Metriq PRISM vector assets used in the design, an exact-rational generator for Figure 1, and a finite consistency checker for the displayed identities and residue bijections. The scripts are supplementary verification aids; the mathematical argument does not depend on numerical computation. No empirical data are used in this work.

Institutional disclaimer and computational research declaration

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Generative artificial intelligence and other computational tools were used as research instruments to support construction search, symbolic and numerical analysis, literature review, proof development, verification, and manuscript preparation. Their use does not constitute independent validation of the results presented.

This manuscript has not yet undergone independent peer review and should not be regarded as an established resolution of the stated problem unless and until its arguments have been examined and confirmed by qualified mathematicians. Independent verification, criticism, replication, and identification of errors are expressly invited.

Creator and contributor statement

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Data and code availability

No external empirical dataset was used. The complete Lua $\text{\LaTeX}$ source, figures, verification code, recorded output, review guide, and release checksums are included in the source and reproducibility package associated with this candidate edition. Computational checks verify only the claims explicitly encoded by those programs and do not independently validate unencoded infinite arguments, interpretation, or novelty.

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Daniel H. Jeffery is the founder and Executive Director of Metriq Foundation, Inc. No other competing interests are declared.

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